

PViT: A Hybrid Model for Deepfake Face Detection using Patch Vision Transformers and Deep Learning

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Abstract—The proliferation of AI-generated deepfakes, particularly facial image forgeries, poses a significant threat to digital security by facilitating misinformation, identity theft, and privacy breaches. Traditional detection approaches, primarily based on Convolutional Neural Networks (CNNs), often exhibit limited effectiveness when confronted with highly refined or subtle manipulations, leading to compromised detection performance. To address this challenge, this study explores the application of Vision Transformers (ViTs), which leverage self-attention mechanisms to capture fine-grained inconsistencies in visual patterns. This research proposed a hybrid deepfake detection model that integrates patch-oriented ViTs with CNN architectures to improve discriminative feature extraction. Experimental evaluation on benchmark datasets demonstrates that the proposed model achieved a detection accuracy 99%, precision 99%, recall 99%, F1-Score 99% on a validation set comprising 76,161 facial images, outperforming conventional CNN-based methods. These results highlight the potential of transformer-based architectures in advancing the robustness and reliability of deepfake detection systems, thereby contributing to the protection of digital authenticity and information integrity.

Index Terms—Deepfake Detection; Vision Transformer (ViT); Deep Learning; CNN; Image Manipulation; Generative Adversarial Networks (GANs); Patches

I. INTRODUCTION

The digital solutions in the era of 5G & 6G technologies rely heavily on Artificial Intelligence (AI), which makes machines capable of autonomous operations through human intelligence simulations. The adversaries existing in the digital world are also relying on AI and misusing it for misinformation and identity theft, in particular. AI tools are used to generate deepfake images, videos, audio, and textual content. Deepfake uses techniques such as Generative Adversarial Networks (GANs) to build synthetic media where an object, especially an individual's face, in samples such as images and video clips, is altered or even replaced. This is also the reason for the continuous evolution of tampering tools because of the fast development of digital image, video, and audio recognition [1] [2]. These tools enable a person with limited resources to produce hyper-realistic graphics easily [3] [4].

The generation of new fake multimedia has become easily available; this is dangerous to the credibility of any digital media and helps spread fake news, identity theft, and cyber-

bullying [5] [6]. Deepfake enables the swapping of one person's face with another person's, something that results in realistic-looking content that poses reputational risks and propagates fake news [7] [8]. In an era where people mostly use illustrations in the form of images, it is essential to guarantee the reliability of digital information. Deepfake technologies improve the use of computer graphics and image processing and turn machine learning algorithms into video mimicry tools that are hard to distinguish. Such techniques include fine texturing, complex modeling, and perfect synchronization of substitution facial expressions and actions producing photorealistic fakes. Some of such developments have made deepfakes a powerful instrument in the hands of infamous elements who can spread fake news, cause fluctuations in stock exchange rates, or slander anyone by producing 'fake' videos of public figures or the general public. Such usage can provoke some serious outcomes like conflict between people and loss of reputation in their community. For example, the first known case of facial transformation can be traced back to images of Abraham Lincoln; this technology has its background [9]. Currently, improvements in the field of ML and artificial neural networks (ANNs) allow reconstruction of the modified image or video to such an extent that it could be easily mistaken for the genuine document [10].

It is important to protect societies from the threats generated by deepfakes in order to preserve trust within society. On the one hand, AI is the driving force behind the generation of deepfakes, while on the other hand, AI has features that are useful in counteracting them. The application of deep learning to identify fake images has emerged as critical, since deepfake media has become a real threat to identities online. New kinds of fake news that imperil society include deepfakes crafted by sophisticated methods such as GANs, creating highly realistic visual content likely to be used for disinformation, identity theft, or to damage someone's reputation. This requires solid detection methods to maintain the confidence of people and the validity of new media outlets.

Deepfake detection by AI is based on the examination of spatial and temporal changes in the images. CNNs that are typically used for image classification tasks have a resolution to these forgeries because of their pixel-level analysis,

for instance, lighting, shadows, or reflections [11]. Sequence models like Recurrent Neural Networks (RNNs) cannot only be used for temporal coherence analysis to look for temporal consistencies of motion and facial expressions. Due to the problems in existing conventional detection methods, which have relatively low detection rates and high error rates, this paper designs a transfer learning-based model to improve deepfake image detection [12] [13].

Current technologies, including GANs, that create deepfakes based on alterations that require a reference picture and desired changes [14] [15]. Moreover, applications such as FaceApp and others have made a clear trend toward the generation of deepfakes more apparent; therefore, promoting the development of automated control systems, as well as combat fake information in general and deepfakes in particular, as they can have critical implications for security and society [16] [17]. Unlike most traditional feedforward neural networks that require the selection of filters by the designer, CNN is capable of learning features of images, which makes it quite suitable when it comes to detecting deepfake artifacts. Appropriate measures for the development of effective detection systems remain a major problem as techniques described as deepfakes become more popular. Using both CNNs and ViTs in a hybrid architecture helps the model identify textures closely and capture far-reaching interactions between features.

This study presents a hybrid model, PViT, to overcome some of the issues with current CNN-based methods and also to make use of the global contextual features of Vision Transformers (ViTs). PViT is designed to (1) detect important features at both local and global parts of images, (2) be robust against small image changes, and (3) generalize well and remain computationally efficient.

The main contributions of this work are as follows:

- 1) This research develops PViT which uses hybrid architectural elements between ViTs and CNNs to track global and local facial irregularities effectively for manipulation identification.
- 2) The model succeeds at spotting filtrated face forgeries and minor manipulations because it investigates distinct texture distributions and spatial structural irregularities.
- 3) The self-attention capability of ViT detects face inconsistencies across all areas better than CNNs do by capturing distant relationships.
- 4) Performance of the proposed PViT Model is evaluated and compared with CNN model where PViT outperforms with 99% detection accuracy.

This paper suggests a new direction of research for ViTs in the multimedia forensics domain, including the ability to uncover manipulations missed by CNNs. Such solutions can scale and become much more efficient for real-world deepfake challenges.

The rest of the paper is organized as follows: In section II a detailed literature review is presented, in Section III a detailed view of the proposed PViT model based on CNNs is presented, Section IV describes the results of all the learning models and comparison, lastly, in Section V a brief conclusion of the study is provided.

II. LITERATURE REVIEW

This section analyzes methods for deepfake detection, categorizing machine learning models like CNNs, transformers, and binary neural networks. The recent work on ViT models and explainable AI for improving detection and multimedia security is presented.

Akhtar and Dasgupta [18] compared different local feature descriptors in order to identify manipulations in deepfake videos with the help of “DeepfakeTIMIT”. The features of noses, cheeks, or jawlines they defined proved useful at spotting these changes, providing a firm base for forensic use in deepfake detection. Bekci et al. [19] introduced a deepfake detection model for cross-dataset, which uses metric learning and steganalysis-rich features. This approach was evaluated on datasets such as FaceForensics++ and CelebDF and led to better accuracy and better performance especially for the manipulations that are hidden in the images.

Li et al. [20] proposed a technique for detecting fake videos developed using AI by focusing on eye blinks. Their result depicted that using the variation in the blinking patterns, researchers can establish a new way to detect whether a video is a deepfake or a real one by considering biological signals. Nguyen et al. [21] considered the eyebrow region analysis as one of the deepfake detection approaches by using deep learning networks including ResNet and SqueezeNet. Their research ascertained high accuracy as a fake content recognition model for specified datasets, to show the applicability of facial feature-based detection.

Ciftci et al. [22] presented one of the first methods for identifying the source of deepfake content by utilizing biological signals within the residuals. Using their approach on various deepfake generators, the participants demonstrated high levels of accuracy, thus pointing to the efficiency of biological signals in determining the source’s truthfulness. Yang et al. [23] proposed MSTA_Net, the model that targets at detecting anomalies of image textures in the whole images instead of the facial areas only. This broader analysis approach enabled enhanced deepfake detection accuracy because it revealed inconsistencies in texture. Wang et al. [24] employed a multi-modal and multi-scale transformer model for improving the deepfake detection process as the transformer model analyzed image patches at multiple receptive fields. This method enhanced the stability and reliability of estimates as well as reducing the errors resulting from global and local disparities of small image patches.

Liu et al. [25] reviewed the use of ViTs in image-analysis tasks and highlighted that the former is beneficial for tasks whose performances depend on global features, such as deepfake detection. Violent Transformers or ViTs efficiently measure the inefficiencies in deepfake content by using self-attention modules. Jeeveswaran et al. [26] presented a detailed analysis of the use of ViTs for dense prediction with clear demonstrations in image segmentation, object detection, and scene parsing. The paper describes how ViTs rely on self-attention mechanisms to surpass the CNNs’ performance in dense

TABLE I
COMPARISON OF EXISTING WORKS ON DEEPPFAKE DETECTION TECHNIQUES

Ref (Year)	Method	Classification Model	Dataset
[28](2024)	Hybrid Vision Transformer and Deep Neural Network	ViT, CAE	Not Specified
[29](2023)	Transfer Learning Approach	CNN	ImageNet, YouTube videos, Celeb-DF
[30](2023)	Improved Dense CNN Architecture	Proposed CNN	7 different datasets (CelebA, FaceForensics++, DeepFakeDetection, iFakeFaceDB, DFDC, CelebDF, Google Jigsaw)
[31](2023)	Deep Learning Approaches	ResNet50	CelebA-HQ, FFHQ
[32] (2023)	Machine Learning	SVM with PCA	-
[33](2024)	Discrimination Method	ResNet-34	CelebA, FFHQ1, ImageNet
[34] (2024)	Real-time Detection	Binary Neural Networks (BNNs)	COCOFake, DFFD, CIFAKE
[35] (2023)	Explainable AI Methods	InceptionResNetV2, DenseNet201, InceptionV3, ResNet152V2	FFHQ
[36] (2023)	Deep Learning	InceptionNet	Not Specified
[37](2023)	Deep Learning	Inception-ResNet-V2	Not Specified
[38] (2022)	Transformer-Based Approach	Transformer	FOFDTD, FF++, DFD, DFDC-P, Celeb-DF
[39] (2022)	Deep Learning	NASNetLarge CNN	Celeb-DFv2
[40] (2022)	Deep Learning	Multi-path CNN	FF++, Celeb-DF, Google DFD

prediction settings, and that is why these models could become effective instruments for image authenticity assessment.

Mary and Edison [27] presented a literature review on deep learning-based approaches for deepfake detection. They reviewed a number of CNN-based and GAN-based methods and also explained that these approaches have various drawbacks; for instance, they may have problems with generalization when trained on new datasets. Some researchers found that the existing models lack the robustness or generalization to offer reliable deepfake detection. Janutėnas et al. [41] on types of image manipulation, including tampering, splicing, and cloning using deep learning. They also underscore the need for pre-trained models and feature extraction as key influencers of the detection capability. It also recognized deficiencies in dealing with high-resolution forgery, which led to hybrid models.

Heidari et al. in [5] surveyed deepfake detection methods and presented a comprehensive taxonomy. They compare the results of different architectures of deep learning such as the LSTMs and autoencoders, on their datasets. In the study, they also emphasized the need to improve the interpretability of deepfake detection systems to help improve public trust in AI systems. Gayathri et al. [42] pointed out some of the limitations to deepfake detection, including scalability to the large dataset and the invention of new forged techniques.

Tankala and Rao proposed robust image falsification detection using a new framework based on ViTs in [43]. They concluded that compared to local relations, ViT is better at identifying global relations of images and accurately detecting forgery, especially for images where subtle changes have been made. Similarly, Tan et al. [44] found that ViTs are active learners in image copy detection, where the model asks for the location of ambiguous areas and improves the detection speed. Iqbal et al. [45] conducted a review that examined the forensic difficulties caused by the introduction of smart devices. During the discussion of evidence tamper detection on various platforms, AI and blockchain technology serve as supporting instruments according to the authors.

Shilaskar et al. [46] put their effort into the detection of GAN-synthesized human face images and emphasized that all GANs retain specific artifacts within the synthesized images that CNNs and ViTs can easily detect. They focused on the real-time applicability of their algorithms for streaming media. Contemporary studies concern themselves with concurrent architectures and self-supervised learning in attempts to enhance existing detection solutions. Gomes et al. [47] used a self-distilled ViT to extract features to detect realistic facial deepfakes, and the main advantage of the model hinges on the model's ability to train the model's discriminative patterns without requiring additional annotations. Due to the

merits of ViT, Shahin and Deriche offered the fusion of ViT learning and deep-neural-network (DNN) learning for even better results since both can capture global features and local features, separately [28]. Although many studies have explored ViT and CNN-based models individually, only a few have combined patch-based ViTs and CNNs in a hybrid architecture specifically designed for deepfake detection. This study specifically aims to fill that gap.

III. PROPOSED PATCH-WISE VISION TRANSFORMER BASED MODEL (PViT)

The intended methodology uses the ViT and CNN variants, VGG16, VGG19, and ResNet50, for feature extraction and classification, respectively. Normalization, as well as resizing or even cropping images, guarantees compatibility with the models in addition to enhancing the model convergence during training. Random flips, rotations, and changes in sharpness of the images increase the richness of the dataset and are used to perform better in different scenarios.

To ensure the performance of the model, the dataset is appropriately divided into a training set, a validation set, and a test set. Regularly, each model is trained using transfer learning, which means that the weights are pre-trained on ImageNet. The prediction problem involves the identification of input images as Fake or Real images. Therefore, accuracy, recall, precision, F1-score, confusion matrix, and ROC-AUC are used while evaluating the models to determine their ability to overcome deepfakes challenges.

A. Dataset Selection

This work employs the “Deepfake and Real Images” dataset downloadable from Kaggle [48], involving an overall of 190,335 RGB images with a dimension of 256 x 256, and the images were sampled as real and fake face images. For our evaluation, we used a validation set comprising 76,161 images, including 38,080 real images and 38,081 fake images. This dataset has been used once in the study by Iqbal et al. [13], and is the base reference study for our work.

The dataset contains visually distinct examples of real and fake faces, as shown in Fig. 1 and Fig. 2.



Fig. 1. Sample of fake images from the “Deepfake and Real Images” dataset.



Fig. 2. Sample of real images from the “Deepfake and Real Images” dataset.

B. Data Preprocessing and Augmentation

To prepare the dataset for training, the following preprocessing and augmentation steps were applied to improve the model’s robustness and generality:

- 1) **Resizing and Cropping:** The size of all images is adjusted to 224x224 pixels according to the input size of the Vision Transformer (ViT) model. Cropping is also performed randomly to address issues of image framing and make the model more robust to spatial changes.
- 2) **Data Augmentation:** A Random Horizontal Flip is applied to the folds to introduce face orientation differences as observed in real life. Random rotation is also applied to introduce distortions, and Random Sharpness Adjustments to make the model more immune to variations. Anti-scale and Center-Cropping are used during validation to ensure that the inputs consistently used for measurement of accuracy were cropped at the center.
- 3) **Normalization:** Each image’s pixel values are normalized using the ImageNet dataset’s mean and standard deviation values:

$$\text{Mean} = [0.485, 0.456, 0.406], \quad \text{Std} = [0.229, 0.224, 0.225]. \quad (1)$$

This guarantees that the pixel intensity values are scaled to what the ViT model needs for improved convergence during training.

C. ViT Architecture

In this study, ViTs are used as a baseline model to detect deepfake images due to their ability to perform complex attention and feature extraction. The given methodology focuses on the subsequent use of ViT components and the identification of integration methods that will increase the accuracy of the detector and its resistance to MNIST’s transformations. In the ViT architecture, images are initially resized to 224x224 in size and are divided into 16x16 sized non-overlapping patches. As a result, each image has a grid of 196 patches, which helps to simplify the network design. ViT-Base is pre-trained on around 14 million images classified into 21,841 different categories from the ImageNet-21k dataset. In total, the architecture consists of 12 transformer encoder layers, each with 768 hidden units, for about 85.8 million parameters. The setup allows enough capacity to model complex visual patterns [28]. The ViT model contains several parts, which are explained in the following (also shown in Fig. 3).

Key Components of the Proposed PViT Model:

- 1) **Patch Embedding:** The input image is segmented into fixed-size patches $(x_1, x_2, \dots, x_N)$, and each patch is transformed into the embedding space by a linear projection matrix (W_{patch}) . The embedding process is defined as:

$$E = [W_{\text{patch}}x_1, W_{\text{patch}}x_2, \dots, W_{\text{patch}}x_N] \quad (2)$$

This step also efficiently transforms the image patches into a sequence for processing with the transformer.

- 2) **Positional Embedding:** To encode the spatial relation between the patches, the positional information $(p_1, p_2, \dots, p_N)$ is added to the respective patch embeddings $(e_1, e_2, \dots, e_N)$. The positional embeddings are added to the patch embeddings as follows:

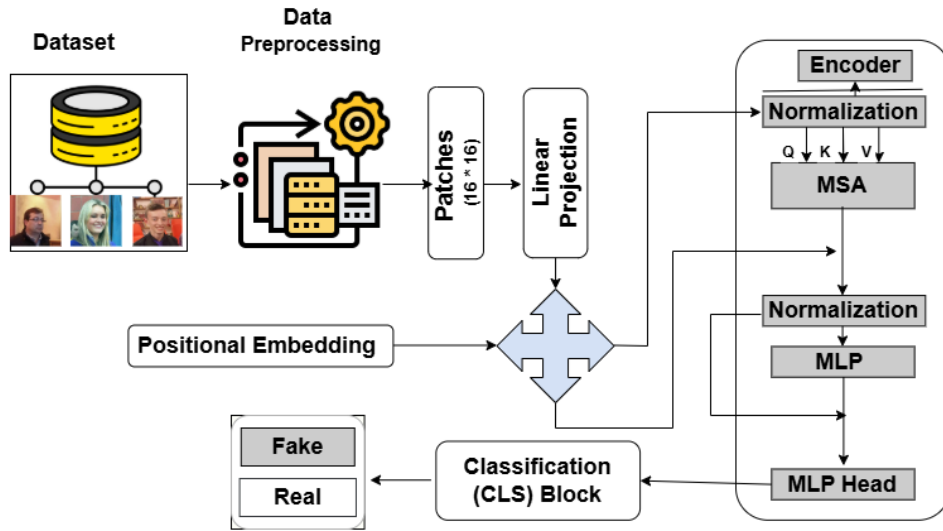


Fig. 3. Block diagram of Proposed patch Vision Transformer model (PViT).

$$E_{POS} = [e_1 + p_1, e_2 + p_2, \dots, e_N + p_N] \quad (3)$$

This augmentation helps the model to incorporate spatial context that is necessary for detecting spatial anomalies that are almost always observed in images of deepfake.

- 3) **Transformer Encoder:** The transformer encoder utilizes positional embeddings (E_{POS}), and self-attention and feedforward neural networks are applied dynamically across layers. The encoder generates contextualized embeddings for each patch, represented as:

$$\text{TransformerEncoder}(E_{POS}) = [z_1, z_2, \dots, z_N] \quad (4)$$

These output embeddings ($z_1, z_2, \dots, z_N$) consist of features and information abstractions of the input image, which embody both local and global connections.

- 4) **Multi-Head Self-Attention (MSA):** The encoder uses MSA to process different parts of an image simultaneously, making detection of manipulations easier. Each attention head independently computes unique representations, which are concatenated to form a comprehensive image representation:

$$\text{MSA}(Q, K, V) = \text{Concat}(H_1, H_2, \dots, H_n) \quad (5)$$

where Q , K , and V stands for the query, key, and value matrices, and $H_1, H_2, \dots, H_n$ denote the attention head outputs [25] [49].

PViT achieves novelty through its design, which brings together the global vision power of ViTs and the local feature extraction strength of CNNs. ViTs are highly effective at relating long-range dependencies across different patches of an image, but CNNs do well at identifying different textures or sharp spots at the edges. Because of this approach, PViT

can identify both easily spotted and hidden fakes in pictures, an ability that has not been looked at as much in previous work.

D. CNN Architecture-based pre-trained models

In this paper, the goal is a binary classification of manipulated and genuine deepfake images using ViT and CNN models. We use VGG16, VGG19, and ResNet50, together with ViT, to differentiate real images from fake ones. These models solve the problem of detecting complex manipulations that are caused by techniques such as Stable Diffusion and StyleGAN2. Thus, the primary objective of this study is to develop deepfake detection using two methods based on CNN and ViT. To evaluate the performance of these models, CNNs architectures are considered initialized with weights learned from the ImageNet dataset to preserve these key features from the source domain. In the fine-tuning process, the weights of the convolution layers and the down-sampling layers of typical CNNs are fixed, and the original fully connected layers are redesigned to fit a binary classification model. In the same way, ViT is precision tuned to add typical global and local representations or features. For the binary classification task, the models add a flattened layer followed by dense layers with two output neurons, which was expected given a binary classification task. To measure the distance of samples from the classes, a SoftMax activation function in the output layer computes class probabilities. This approach leverages the CNNs' ability to detect small manipulations because of their local feature extraction, to which ViT offered a broader understanding of all details and context. Therefore, this study discusses the integration of ViT in utilizing the latest and traditional patches based on modern and traditional CNNs for improved deepfake detection performance. Despite being outperformed on test instances, CNNs, which accurately detect manipulated areas, cannot be replaced, whereas ViT

complements the detection model by examining the image for structure and observing the global and local manipulations.

E. Evaluation Metric

The model performance is assessed using several key metrics, each reflecting the ability of ViT and CNN-based models to accurately classify deepfakes:

- Accuracy: Represents the proportion of correctly classified images, calculated as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (6)$$

where TP denotes true positives, TN represents true negatives, FP refers false positives, and FN = false negatives [50].

- Precision: Calculated by dividing the true positives by the sum of the true positives and false positives. It reflects the model's ability to minimize false detections:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (7)$$

This ensures that any identified deepfake includes minimal false alarms, avoiding incorrect results [51].

- Recall: It is the proportion of true positives to the added true positives and false negatives. Recall reflects the model's sensitivity:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (8)$$

High recall is essential to identify the majority of deepfakes, as noted in [13].

- F1 score: Represents the harmonic mean of precision and recall, promoting a balance between the two metrics:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

The F1 score is especially beneficial when precision and recall are equally important [28].

- ROC-AUC score: Measures the model's ability to discriminate or separate between classes at various threshold levels. A high ROC-AUC score signifies the strong capacity of the model in classifying real and fake images [52].

IV. EXPERIMENT RESULTS AND DISCUSSION

In this work, the ViT-based model and typical CNN models are examined for deepfake identification. The results indicate a significant advantage for ViT in performance and are explained by the program's capacity to incorporate global and local components using self-attention techniques. In the following section, a comprehensive report of the outcome assessment is provided.

A. Performance Evaluation of PViT and CNNs

The baseline model, VGG16, achieves an accuracy of 96%, a precision of 96%, along with comparable metrics. Although it is efficient in identifying manipulations, its system architecture focuses only on the extraction of localized features, making it less generalizable to high-dimensional synthetic datasets.

VGG19 slightly improves with an accuracy of 97% due to additional layers that detect finer details of an image. However, this improvement comes at the cost of relying on localized patterns, leading to performance saturation.

ResNet-50 exhibited comparable performance, maintaining a uniform accuracy of 96%. This performance is attributed to residual connections that address vanishing-gradient problems. However, the absence of attention mechanisms limits its ability to learn global discrepancies in intricate, detailed, deepfake content.

TABLE II
PERFORMANCE OF PRE-TRAINED MODELS ON THE GIVEN DATASET

Model	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	ROC (%)
VGG19	97	97	97	97	97
VGG16	96	96	96	96	96
ResNet-50	96	96	96	96	96

B. Patch Vision Transformer (PViT) Analysis

The ViT model demonstrates substantial improvement, achieving an accuracy, recall, precision and F1-score of 99% compared to CNN-based models. The self-attention mechanism of the ViT enables it to process images in different patches without neglecting smaller details, which are often overlooked by other models. This observation aligns with the findings of [53], who emphasized that ViT effectively applies global context in tasks such as deepfake detection.

TABLE III
PERFORMANCE OF THE PROPOSED PViT MODEL.

Class	Precision (%)	Recall (%)	F1-score (%)	Support
Real	99.14	99.35	99.24	38080
Fake	99.34	99.14	99.24	38081
Accuracy	-	-	99.24	76161
Macro_avg	99.24	99.24	99.24	76161
Weighted_avg	99.24	99.24	99.24	76161

The training and validation accuracy and loss curves, presented in Fig. 4, illustrate the stability of the model and the superior learning capacity. The accuracy graph shows a steady improvement, reaching nearly perfect values by the final epoch. Concurrently, the loss graph demonstrates a steady decline in both training and validation loss with minimal fluctuations, underscoring the ViT model's efficiency in binary classification tasks.

The ROC curve, shown in Fig. 5, highlights the high discriminatory power of the ViT model with an AUC close to 1. This demonstrates the model's ability to distinguish between real and fake images across varying threshold values, further

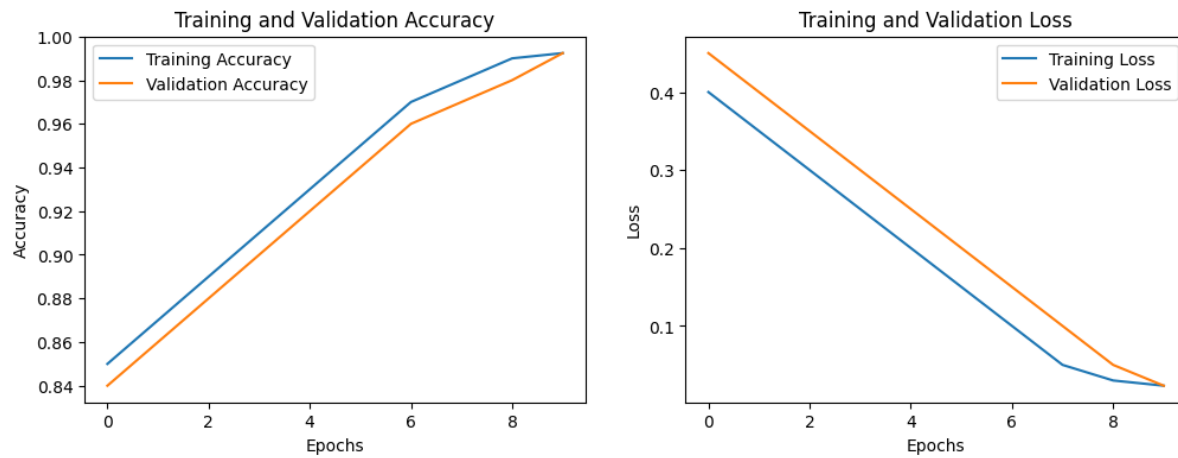


Fig. 4. Training and Validation Accuracy and Loss Graphs for the ViT Model.

affirming its superior performance metrics.

To evaluate the practical efficiency of the PViT model, we measured its computational complexity. The model has 86 million parameters and takes an average of 38 milliseconds to process one image using an NVIDIA RTX 3090. It shows that this dataset might be useful for real-time purposes of detecting deepfakes. Another aspect of ViT's classification performance is evident in the confusion matrix shown in Fig. 6. The confusion matrix reveals a high count of true positives (37,754 fake images correctly classified) and true negatives (37,831 real images correctly classified). The small number of false positives and false negatives reflects the model's robustness in avoiding misclassifications. These results underscore the practical applicability of the ViT model in deepfake detection tasks. Its ability to maintain a balance between true-positive and true-negative values, combined with its remarkable metrics, makes it an effective and reliable approach for such binary classification tasks.

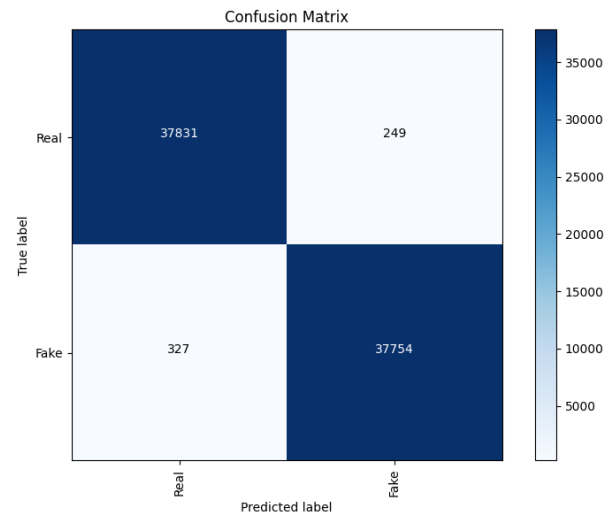


Fig. 6. Confusion Matrix for the ViT Model.

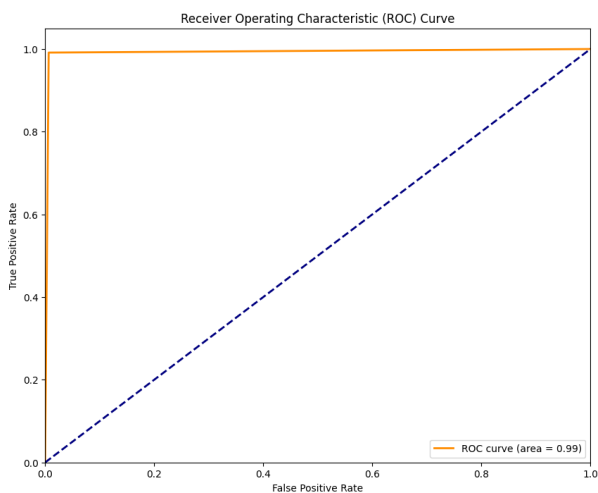


Fig. 5. ROC Curve with AUC close to 1 for the ViT Model.

C. Comparative Results

Our results extend the study of [13], in which the authors used CNN models such as VGG16, VGG19, and the InceptionV3 model for deepfake detection with a maximum accuracy of 90%. Using PViT, this work demonstrates a nine percent increase in performance with accuracy 99%, providing evidence of the efficacy of attention-based architectures in deepfake identification. Both studies utilize the same Kaggle dataset, ensuring comparability of the results obtained, as shown in Table IV.

That ViT achieves such strong performance highlights the method's advantage of processing image patches in their holistic form, enabling the detection of manipulations across both local and global cues. This approach addresses several shortcomings of CNN architectures while reaffirming the importance of transformer models in multimedia forensics.

TABLE IV
COMPARATIVE RESULTS OF PROPOSED PViT MODEL

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
VGG16 [13]	90.00	91.00	90.00	90.00
Proposed (ViT)	99.24	99.24	99.24	99.24

V. CONCLUSIONS

This work demonstrates that PViT is highly effective in the detection of deepfake images, giving 99% detection accuracy. The self-attention processing of ViT surpasses the traditional CNN networks VGG16, VGG19, and ResNet-50. The model becomes better equipped to differentiate between authentic images and tampered images. Transformer-based models are effective for AI-generated deepfakes. More research and development should be devoted to improving the performance of ViT in terms of the number of computations for real-time use and the expansion of the range of multimedia fields in which it can be used. This work can be extended to detect deepfake videos and perform multimodal analysis.

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